

LAB EXPERIMENT RECORD

SET 14 — Survey Results Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	AKILA S
Register Number	192424331
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A. Aim

To analyze survey results by creating a stacked bar chart, radar chart, and survey data table using R, and to develop an interactive Tableau dashboard combining all three visualizations.

B. Dataset

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D

3	C	A	B
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Program 1 — Stacked Bar Chart for Question 1

Aim

To create a stacked bar chart showing the distribution of answers for Question 1.

R Code

```
# Survey Results
```

```
respondent <- c(1, 2, 3)
```

```
question1 <- c("A", "B", "C")
```

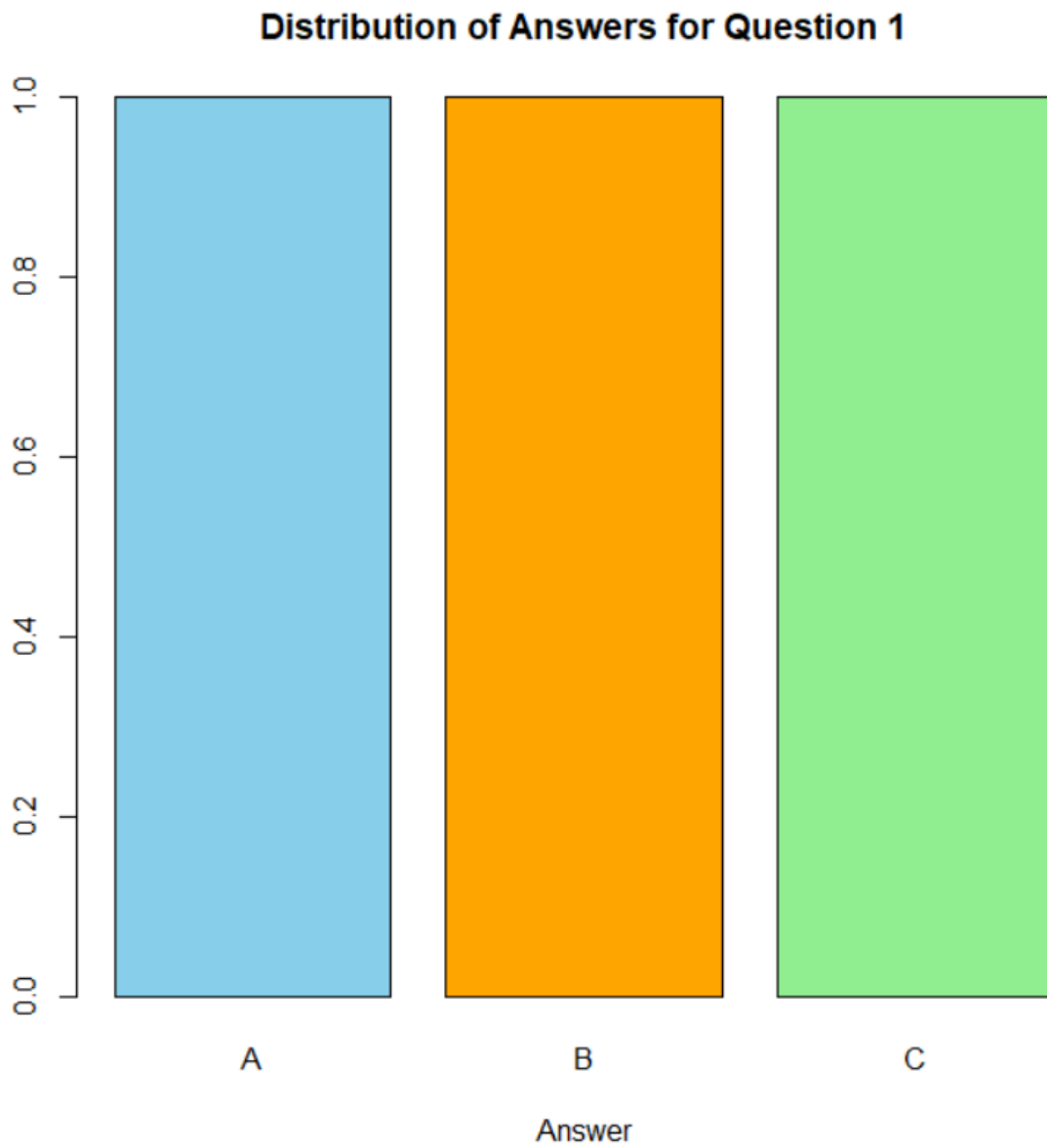
```
survey <- data.frame(
  Respondent = respondent,
  Question1 = question1
)
```

```
# Create frequency table
```

```
response_count <- table(survey$Question1)
```

```
# Create bar chart
```

```
barplot(
  response_count,
  main = "Distribution of Answers for Question 1",
  xlab = "Answer",
  ylab = "Number of Responses",
  col = c("skyblue", "orange", "lightgreen")
)
```



Observation

Answer A, answer B, and answer C each received one response. Each answer represents one-third of the total responses.

Program 2 — Radar Chart for All Three Questions

Aim

To create a radar chart showing the overall pattern of responses across all three questions.

R Code

```
# Install package if required
# install.packages("fmsb")

library(fmsb)

# Survey data
survey <- data.frame(
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Count responses for each answer
answers <- c("A", "B", "C", "D")

response_counts <- rbind(
  Q1 = as.numeric(table(factor(
    survey$Question1,
    levels = answers
  ))),
  Q2 = as.numeric(table(factor(
    survey$Question2,
    levels = answers
  ))),
```

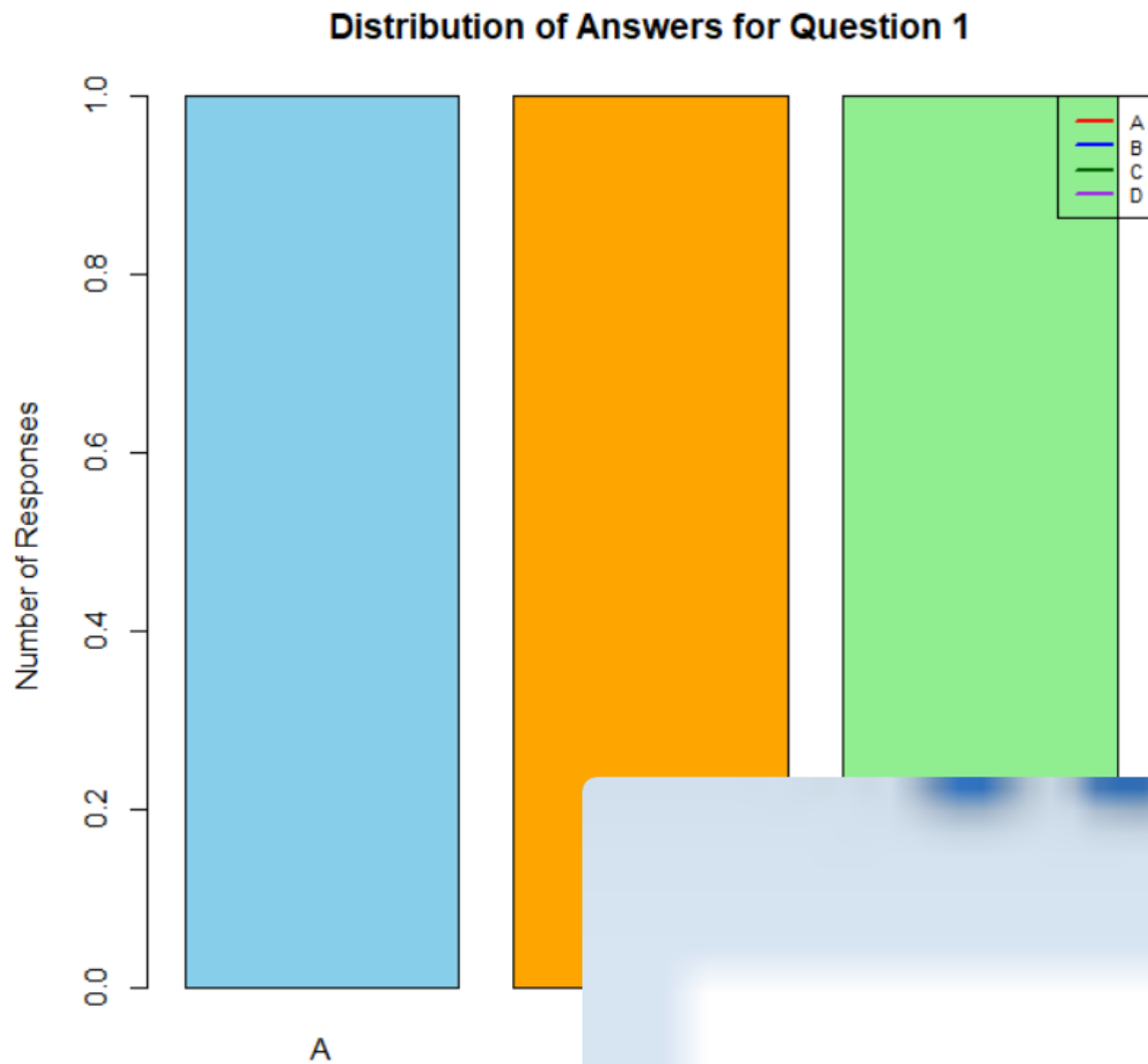
```
Q3 = as.numeric(table(factor(
  survey$Question3,
  levels = answers
)))
)
```

```
# Prepare data for radar chart
radar_data <- as.data.frame(t(response_counts))
rownames(radar_data) <- answers
colnames(radar_data) <- c(
  "Question 1",
  "Question 2",
  "Question 3"
)
```

```
# Add maximum and minimum rows
radar_data <- rbind(
  Maximum = rep(3, 3),
  Minimum = rep(0, 3),
  radar_data
)
```

```
# Create radar chart
radarchart(
  radar_data,
  axistype = 1,
  title = "Overall Survey Response Pattern",
  pcol = c("red", "blue", "darkgreen", "purple"),
  plwd = 2,
  plty = 1
)
```

```
legend(  
  "topright",  
  legend = rownames(radar_data)[3:6],  
  col = c("red", "blue", "darkgreen", "purple"),  
  lty = 1,  
  lwd = 2,  
  cex = 0.7  
)
```



Observation

The radar chart compares the frequencies of answers A, B, C, and D across the three questions. Answer A appears twice in Question 2, while the other answer frequencies vary across questions.

Program 3 — Survey Response Data Table

Aim

To create a table displaying the responses given by each respondent.

R Code

```
# Survey Results Dataset
```

```
survey <- data.frame(  
  Respondent = c(1, 2, 3),  
  Question1 = c("A", "B", "C"),  
  Question2 = c("B", "A", "A"),  
  Question3 = c("C", "D", "B")  
)
```

```
# Display survey table
```

```
print(survey)
```

```
# Display survey table  
print(survey)  
Respondent Question1 Question2 Question3  
      1         A         B         C  
      2         B         A         D  
      3         C         A         B
```

```
|
```

Observation

The table displays each respondent's answers to Questions 1, 2, and 3.

Program 4 — Tableau Interactive Dashboard

Aim

To create a Tableau dashboard combining the stacked bar chart, radar chart, and survey response table for interactive exploration.

Step 1 — Prepare the Excel Dataset

Enter the following data in Excel and save it as Survey_Results.xlsx.

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

Open Tableau and connect to the Excel file.

Step 2 — Create the Stacked Bar Chart

1. Open a new worksheet.
2. Rename it Stacked Bar Chart.
3. Drag Question 1 to Columns.
4. Drag Respondent to Rows and change its aggregation to Count if necessary.
5. Drag Question 1 to Color.
6. Select Bar from the Marks card.
7. Enable Show Mark Labels if needed.

The chart displays the response distribution for Question 1.

Step 3 — Prepare Data for the Radar Chart

The survey data must be converted from wide format to a long format.

1. Open the Data Source tab.
2. Select the three columns: Question 1, Question 2, and Question 3.
3. Right-click and select Pivot.
4. Rename Pivot Field Names to Question.
5. Rename Pivot Field Values to Answer.
6. Create a worksheet named Radar Chart.

Important: Tableau does not provide a standard radar chart directly in every version. A radar chart may require calculated fields and a polygon/path setup, or an extension. If your class requires a native Tableau-only solution, confirm which approach your Tableau version supports before proceeding.

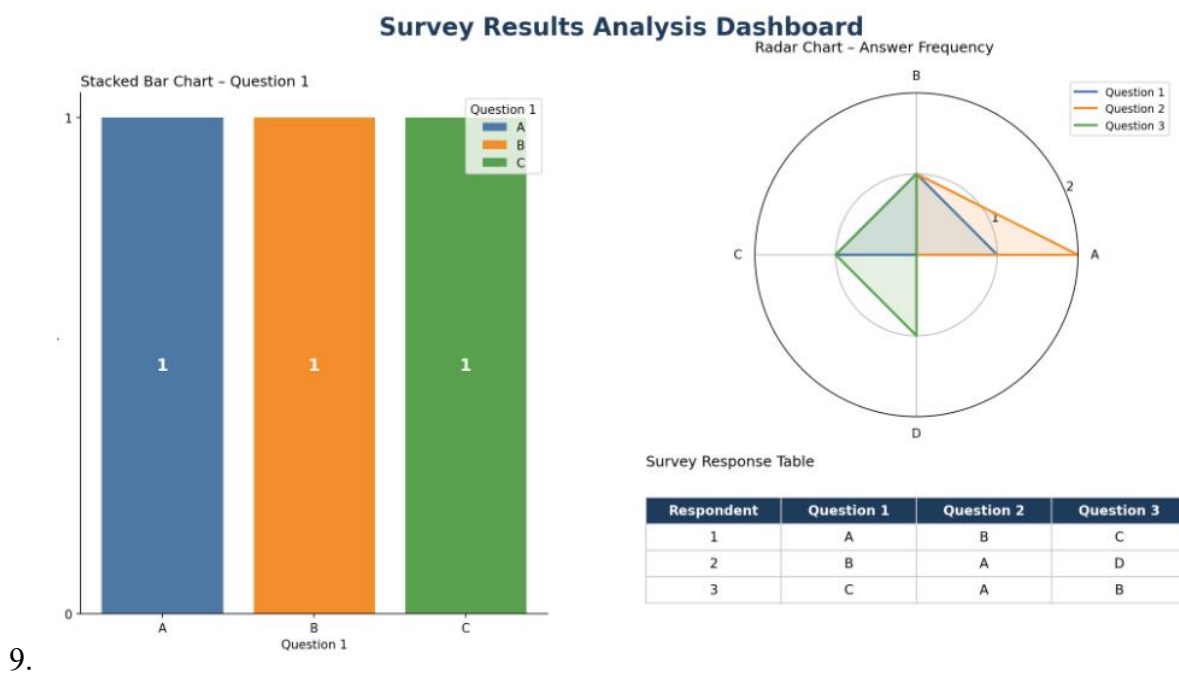
Step 4 — Create the Survey Table

1. Create a new worksheet.
2. Rename it Survey Response Table.
3. Drag Respondent to Rows.
4. Drag Question 1, Question 2, and Question 3 to the view using Measure Names/Measure Values or the text table layout available in your Tableau version.
5. Ensure all three answers appear as separate columns.

Step 5 — Create the Dashboard

1. Click Dashboard → New Dashboard.
2. Set the dashboard size.
3. Drag Stacked Bar Chart onto the dashboard.
4. Drag Radar Chart onto the dashboard.
5. Drag Survey Response Table onto the dashboard.
6. Arrange the three sheets neatly.

7. Add the title Survey Results Analysis Dashboard.
8. Save the Tableau workbook.



Observation

The dashboard combines the response distribution chart, radar visualization, and survey table, enabling users to examine the survey answers in one place.

C. Overall Procedure

1. Create the survey dataset.
2. Generate the stacked bar chart for Question 1 using R.

3. Count answer frequencies for all three questions.
4. Create a radar chart using the fmsb package.
5. Display the respondent data in a table.
6. Import the dataset into Tableau.
7. Create the stacked bar chart and survey table.
8. Prepare the survey data for the radar chart.
9. Combine the worksheets into a Tableau dashboard.
10. Save and review the dashboard.

D. Result

The survey results were analyzed using R, and the stacked bar chart, radar chart, and response table were created. The Tableau dashboard brings these visualizations together for convenient exploration of survey responses.

E. Key Functions Used

Function	Purpose
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data.frame()	Creates the survey dataset
table()	Counts answer frequencies
factor()	Sets answer categories
barplot()	Creates the bar chart
library(fmsb)	Loads the radar chart package
radarchart()	Creates the radar chart
print()	Displays the survey table
pivot in Tableau	Converts question columns into rows
Dashboard → New Dashboard	Creates the Tableau dashboard